Plumb

A peg module for MUSD, built from Mezo's own pool, gauge and savings vault.

Stablecoins become standing bids for MUSD just under its price. What they buy earns in sMUSD and is sold back only at $1 or more.

The book today

Mezo's main MUSD/mUSDC pool, read from chain on Oct 26

5.3M

MUSD offered between $0.99 and $1.00

$38,581

of bids

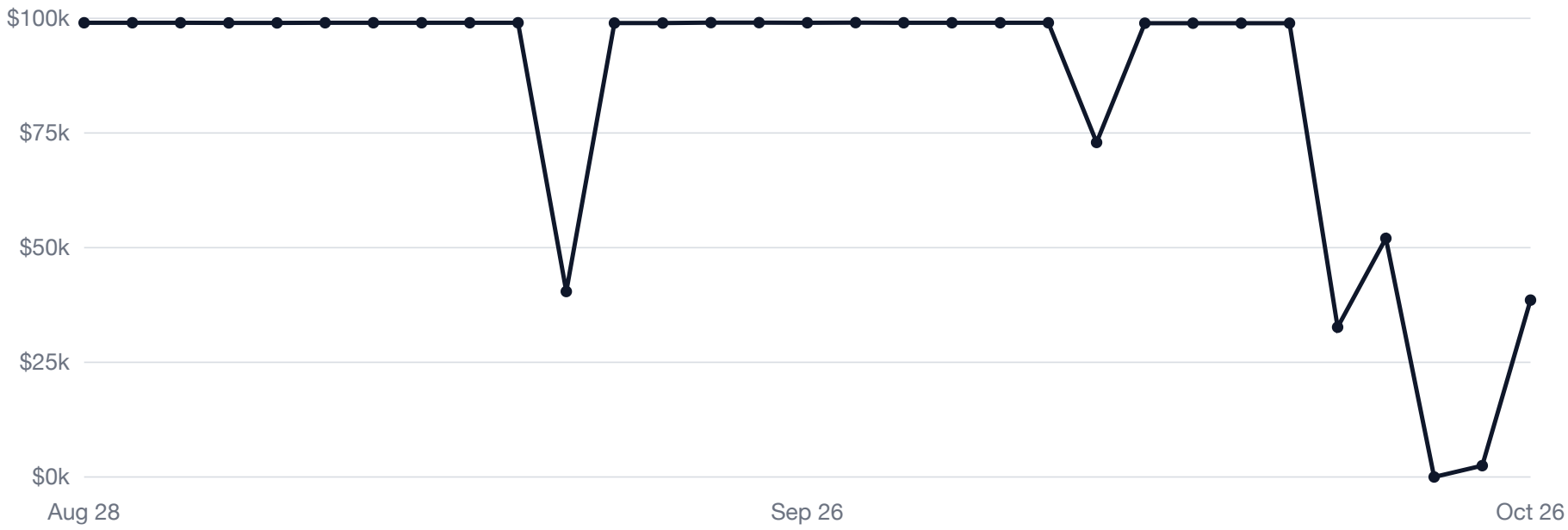
$0.990

MUSD price

Selling 100,000 MUSD into that pool pays $38,556. The rest of the sale finds no buyers.

October

USD paid for selling 100,000 MUSD on Mezo, replayed at each block



Mostly near $99,000 from Aug 28 to Oct 16, with two brief dips. Replayed at a block on Oct 22, a 10,000 MUSD sale would have paid $11.

Why it matters

- The borrowing book holds 14,489 MUSD of debt against 16.59 BTC (Mezo's H1 report: $13.1M of loans in June).
- 96.7% of MUSD supply sits in protocol contracts: Stability Pool, sMUSD, the pool's sell side, the bridge.
- Mezo already pays MEZO emissions for MUSD liquidity. Plumb points that liquidity at the side that is missing.

The vault

PlumbBidVault

1. Deposit mUSDC, get shares.
2. The vault bids from one tick under the market to 2% deeper, staked in the pool's gauge.
3. MUSD it buys earns in sMUSD while MUSD is under $1.
4. Near $1 it is offered back, never below $1.
5. MEZO streams to depositors. 10% is posted back to the same gauge to pull veBTC votes.
6. Exits are pro rata and in kind, any time.

Proof

Mezo mainnet's real contracts on a fork, and live on testnet

$38,556 → $63,556

a 100k MUSD sale, before and after a $25k deposit (fork)

$0 → $389

a 400 MUSD sale, block before and after our testnet deposit

0 MEZO

to a flash-loan deposit-claim-redeem attack

sMUSD strategy

PlumbAskStrategy

- sMUSD holds 7.38M MUSD in its default IdleStrategy.
- One `setStrategy` call: 95% stays idle, 5% becomes a standing offer at $1.001 and up.
- On the fork, the real owner switches all 7,382,555 MUSD over. A full cycle returns 164 MUSD to savers, and switching back returns every token.
- 90% of savers can leave at once with the order filled. Sandwiched buybacks are refused.

Safety

- No owner, no admin, no upgrades, no signatures or meta-transactions.
- Spot must agree with the TWAP before any liquidity moves or any deposit is priced.
- Floor at $0.96 and a 60% inventory cap stop the vault buying all the way down.
- 45 fork tests against Mezo mainnet state, 9 unit tests and 10 invariants. The invariants found a share-pricing bug and two blind reviews found more; each fix has its own test.

Built on Mezo

MUSD/mUSDC CL pool, NFT manager	bids and asks
CLGauge, PoolsVoter, gauge voting reward	MEZO to depositors, incentives back to the peg pool
sMUSD and its strategy hook	parked inventory; the ask strategy
CL router	strategy buybacks
ActivePool, Stability Pool, NTT manager	the public peg observatory

Next

1. Mainnet vault with a deposit cap, seeded by whoever wants MUSD at $1: the foundation, market makers, borrowers.
2. The same vault on Aero's MUSD/USDC pool on Base, where Mezo's gauges already pay.
3. A governance proposal, with the fork test as evidence, to put sMUSD's idle MUSD to work under published caps.